

# COURSE PROJECT REPORT

EE624: Speech Technology

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## Project Titles

**Project 1:** DTW-Based Isolated Vowel Recognition

**Project 2:** GMM-Based Isolated Digit Recognition

**Project 3:** GMM-HMM and DNN-HMM Based ASR Systems

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# 1 Introduction

This report documents the implementation of three speech recognition projects for the course EE624: Speech Technology. The objective of each project is to apply classical and modern speech processing techniques to build isolated and continuous speech recognisers.

- **Project 1** applies Dynamic Time Warping (DTW) for template-based isolated vowel recognition from self-recorded speech.
- **Project 2** applies Gaussian Mixture Models (GMM) with 16 and 32 components for isolated digit recognition, comparing both systems.
- **Project 3** develops a full automatic speech recognition (ASR) pipeline using the Kaldi toolkit, covering GMM-HMM and DNN-HMM systems for continuous speech.

All recordings were made in the speaker's own voice using Audacity at 16 kHz, 16-bit PCM format, across two sessions to capture natural session variation.

## 2 Feature Extraction: 39-Dimensional MFCC

All three projects use Mel-Frequency Cepstral Coefficients (MFCC) as the acoustic front-end. The extraction pipeline is as follows:

1. **Pre-emphasis:**  $\hat{s}[n] = s[n] - 0.97 \cdot s[n - 1]$
2. **Silence trimming:** Energy-based removal of leading/trailing silence.
3. **Framing and windowing:** Hamming window applied to short overlapping frames.
4. **Mel filterbank:** 26 triangular filters on the Mel scale applied to the power spectrum.
5. **DCT:** 13 base MFCC coefficients extracted per frame.
6. **Delta features:**

$$d_t = \frac{\sum_{n=1}^N n (c_{t+n} - c_{t-n})}{2 \sum_{n=1}^N n^2} \quad (1)$$

7. **Concatenation:**  $\mathbf{x}_t = [\mathbf{c}_t, \Delta\mathbf{c}_t, \Delta\Delta\mathbf{c}_t] \in \mathbb{R}^{39}$

Table 1: Recording and Feature Parameters

| Parameter               | Value                                  |
|-------------------------|----------------------------------------|
| Sampling rate           | 16,000 Hz                              |
| Bit depth               | 16-bit PCM                             |
| Base MFCC coefficients  | 13                                     |
| Total feature dimension | 39 (base + $\Delta$ + $\Delta\Delta$ ) |
| Utterances per class    | 25                                     |
| Recording sessions      | 2                                      |

## 3 Project 1: DTW-Based Isolated Vowel Recognition

### 3.1 Objective

To recognise isolated English vowels (A, E, I, O, U) from self-recorded speech using Dynamic Time Warping as a template-matching classifier.

### 3.2 Methodology

#### 3.2.1 Dynamic Time Warping

Two utterances of the same vowel naturally vary in duration. DTW finds the optimal non-linear alignment between two temporal sequences by computing an accumulated cost matrix:

$$D(i, j) = d(\mathbf{x}_i, \mathbf{y}_j) + \min[D(i-1, j), D(i, j-1), D(i-1, j-1)] \quad (2)$$

where  $d(\mathbf{x}_i, \mathbf{y}_j)$  is the Euclidean distance between frames  $i$  and  $j$ . The normalised DTW distance is:

$$\text{DTW}(\mathbf{X}, \mathbf{Y}) = \frac{D(M, N)}{|\mathcal{P}|} \quad (3)$$

where  $|\mathcal{P}|$  is the length of the optimal warping path.

#### 3.2.2 Recognition Pipeline

1. Extract 39-dim MFCC features from all 125 recordings (5 vowels  $\times$  25 utterances).
2. Select the first utterance of each vowel as the **reference template**.
3. Compute normalised DTW distance from each test utterance to all 5 templates.
4. Classify by minimum DTW distance (1-Nearest Neighbour):

$$\hat{v} = \arg \min_{v \in \{A, E, I, O, U\}} \text{DTW}(\mathbf{X}_{\text{test}}, \mathbf{X}_{\text{ref}}^{(v)}) \quad (4)$$

### 3.3 DTW Score Analysis

Table 2 shows that same-vowel pairs (similar) yield significantly lower normalised DTW distances than cross-vowel pairs (dissimilar), confirming the discriminative ability of the DTW measure.

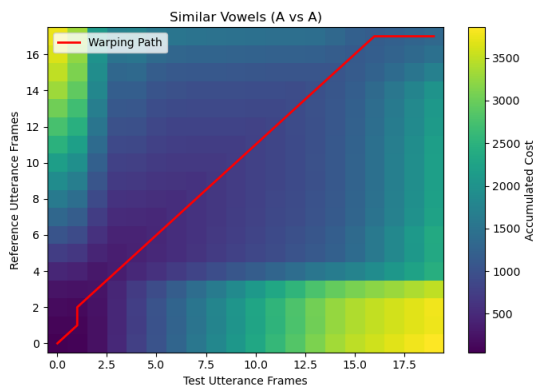
Table 2: Normalised DTW Scores: Similar vs. Dissimilar Pairs

| Pair                          | Type       | Norm. DTW Score |
|-------------------------------|------------|-----------------|
| a vs a                        | Similar    | 63.635          |
| e vs e                        | Similar    | 58.073          |
| i vs i                        | Similar    | 73.945          |
| o vs o                        | Similar    | 52.965          |
| u vs u                        | Similar    | 36.494          |
| <b>Dissimilar Vowel Pairs</b> |            |                 |
| a vs e                        | Dissimilar | 82.586          |
| a vs i                        | Dissimilar | 92.991          |
| a vs o                        | Dissimilar | 115.564         |
| a vs u                        | Dissimilar | 116.376         |

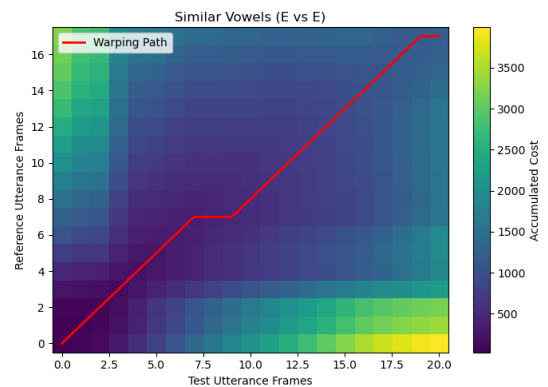
### 3.4 DTW Curve

The DTW curve shows the accumulated cost matrix with the optimal warping path. If the path stays close to the diagonal, the two signals are well aligned and similar. If the path moves far from the diagonal, the signals are poorly aligned and dissimilar.

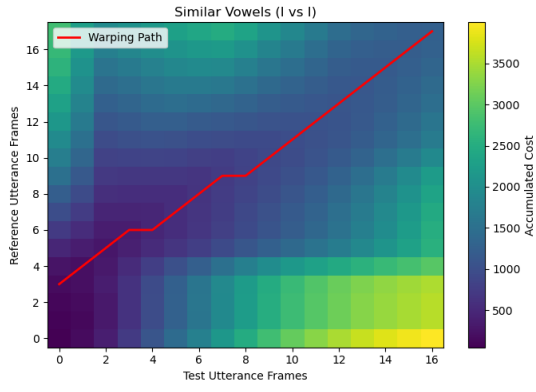
#### 3.4.1 Similar Utterances



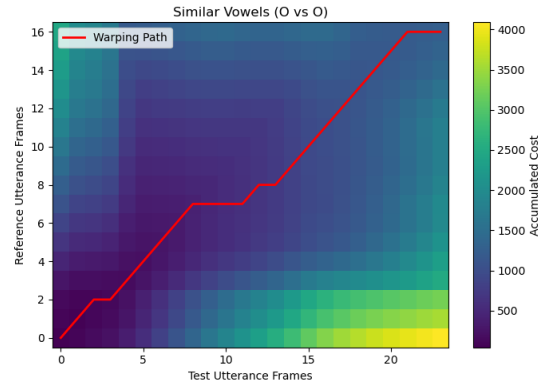
(a) a vs a



(b) e vs e



(a) i vs i



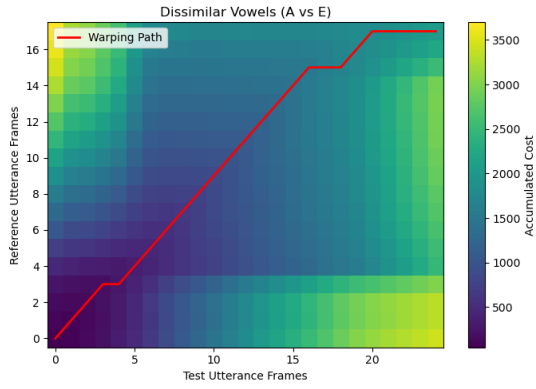
(b) o vs o



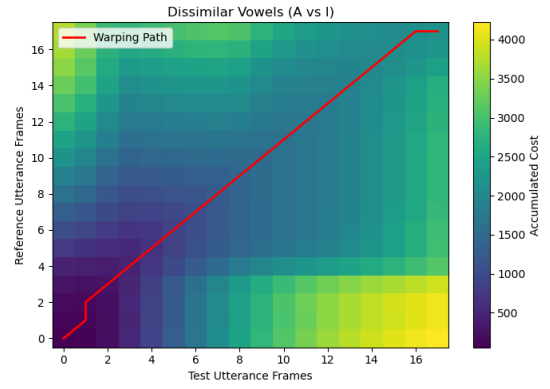
(c) u vs u

Figure 2: DTW alignment plots for similar vowel pairs.

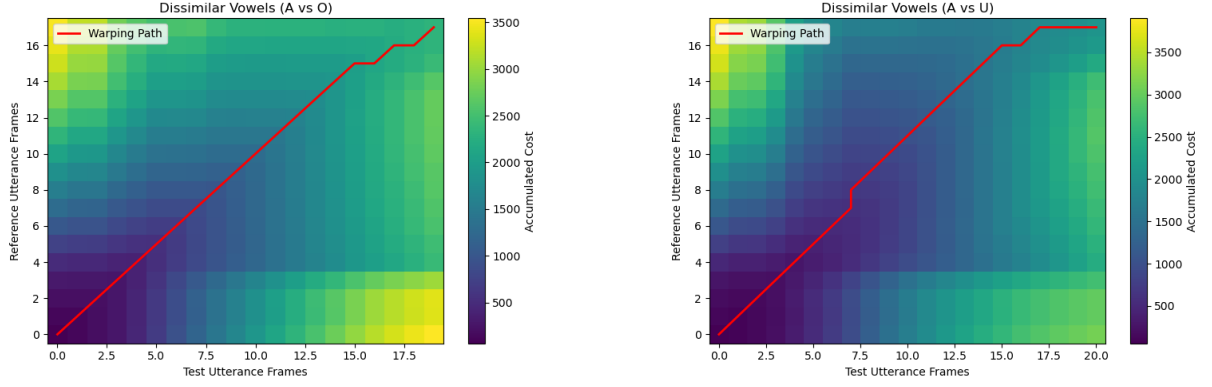
### 3.4.2 Dissimilar Utterances



(a) a vs e



(b) a vs i



(a) a vs o

(b) a vs u

Figure 4: DTW alignment plots for dissimilar vowel pairs.

### 3.5 Results

#### 3.5.1 Confusion Matrix

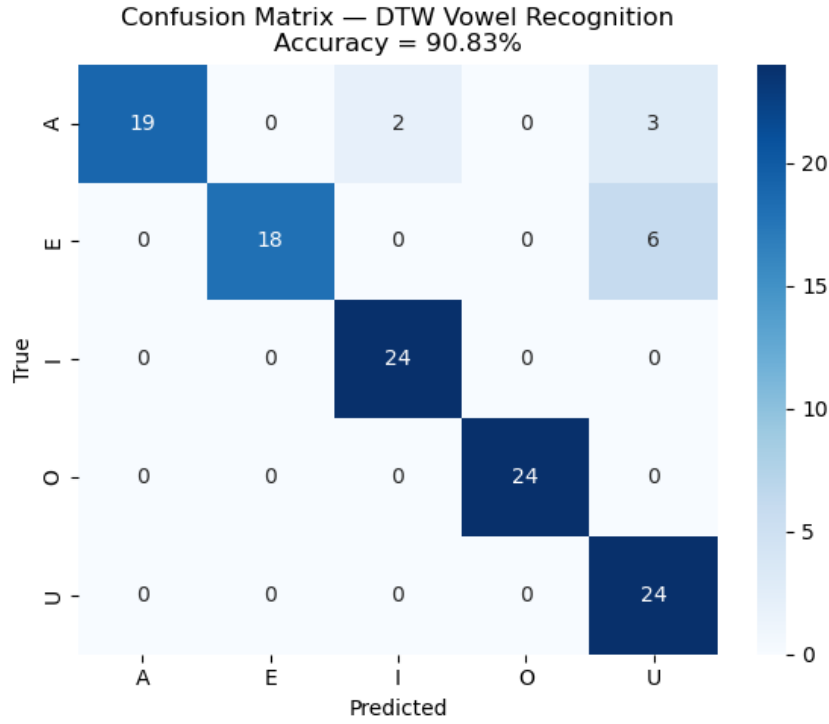


Figure 5: Confusion Matrix — DTW Vowel Recognition

#### 3.5.2 Recognition Accuracy

$$\text{Accuracy} = \frac{\text{Correctly classified}}{\text{Total test utterances}} = \frac{109}{120} \times 100\% = \mathbf{90.83\%} \quad (5)$$

### 3.6 Discussion

- The overall DTW recognition accuracy achieved is 90.83%, which shows that the system performs well for isolated vowel recognition.
- Vowels I, O, and U were recognized perfectly, each with 24 correct classifications and no misclassifications.
- Vowel A was mostly recognized correctly (19 out of 24), but a few samples were confused with I (2 cases) and U (3 cases).
- Vowel E showed the highest confusion, with 6 samples misclassified as U, resulting in 18 correct recognitions.
- The confusion mainly occurs between acoustically similar vowels, indicating that DTW captures similarity in speech patterns but may struggle when vowel characteristics overlap.

## 4 Project 2: GMM-Based Isolated Digit Recognition

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### 4.1 Objective

To recognise isolated English digits (0–9) using Gaussian Mixture Models trained on MFCC features, and to compare recognition performance with 16 and 32 mixture components.

### 4.2 Methodology

#### 4.2.1 Gaussian Mixture Model

A GMM with  $M$  components models the probability density of MFCC features for digit  $d$  as:

$$p(\mathbf{x} \mid \lambda_d) = \sum_{k=1}^M w_k \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \quad (6)$$

where  $w_k$  are mixture weights ( $\sum_k w_k = 1$ ),  $\boldsymbol{\mu}_k$  is the mean, and  $\boldsymbol{\Sigma}_k$  is a diagonal covariance matrix. Parameters are estimated by the EM algorithm.

#### 4.2.2 Recognition Pipeline

1. **Split:** 70% training, 30% testing per digit class (random seed fixed for fair comparison).
2. **Train:** Pool all MFCC frames from training recordings and fit one GMM per digit.

3. **Decode:** Maximum log-likelihood classification:

$$\hat{d} = \arg \max_{d \in \{0, \dots, 9\}} \sum_{t=1}^T \log p(\mathbf{x}_t \mid \lambda_d) \quad (7)$$

4. **Evaluate:** Confusion matrix, accuracy, precision, recall, F1-score.

Table 3: GMM Experimental Configuration

| Parameter               | Value         |
|-------------------------|---------------|
| Feature dimension       | 39            |
| Covariance type         | Diagonal      |
| EM iterations           | 200           |
| EM initialisations      | 3 (best kept) |
| Train / Test split      | 70% / 30%     |
| Mixture counts compared | 16 and 32     |

## 4.3 Results

### 4.3.1 Confusion Matrix — GMM-16

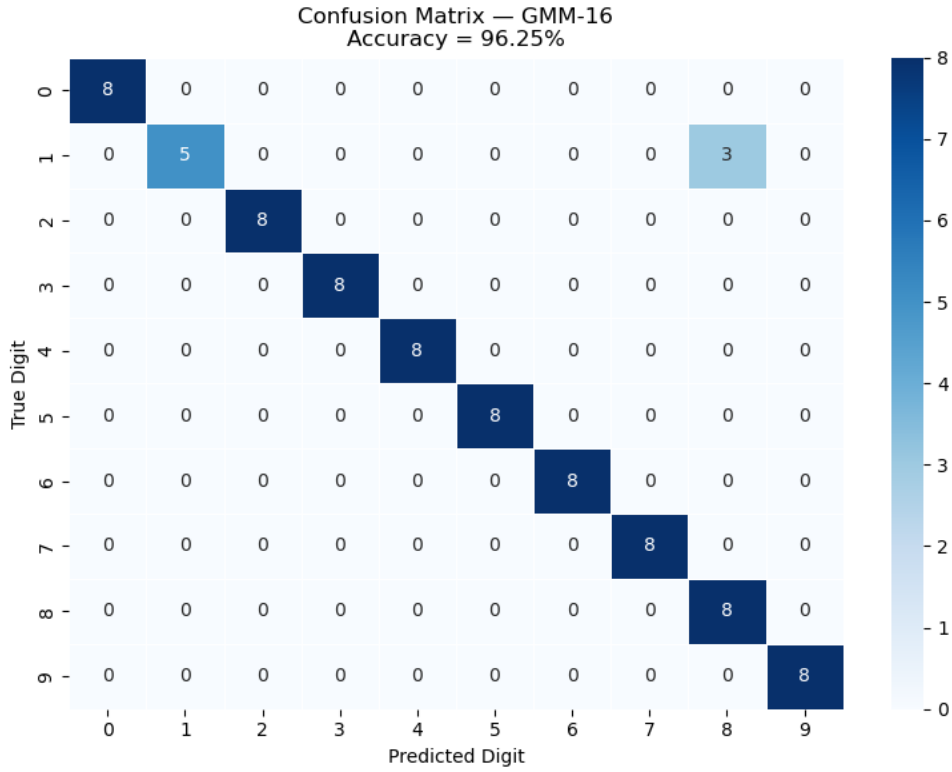


Figure 6

### 4.3.2 Confusion Matrix — GMM-32

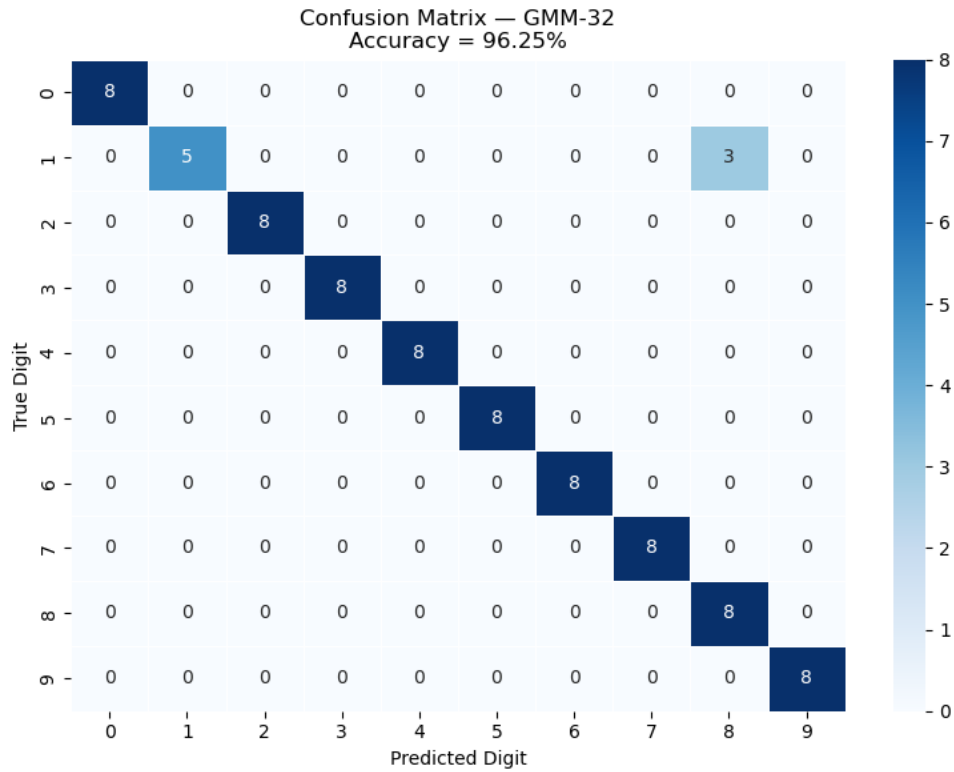


Figure 7

### 4.3.3 Per-Digit Metrics

Table 4: Per-Digit Precision, Recall, and F1-Score

| Digit     | GMM-16    |        |          | GMM-32    |        |          |
|-----------|-----------|--------|----------|-----------|--------|----------|
|           | Precision | Recall | F1-Score | Precision | Recall | F1-Score |
| 0         | 1.0000    | 1.0000 | 1.0000   | 1.0000    | 1.0000 | 1.0000   |
| 1         | 1.0000    | 0.6250 | 0.7692   | 1.0000    | 0.6250 | 0.7692   |
| 2         | 1.0000    | 1.0000 | 1.0000   | 1.0000    | 1.0000 | 1.0000   |
| 3         | 1.0000    | 1.0000 | 1.0000   | 1.0000    | 1.0000 | 1.0000   |
| 4         | 1.0000    | 1.0000 | 1.0000   | 1.0000    | 1.0000 | 1.0000   |
| 5         | 1.0000    | 1.0000 | 1.0000   | 1.0000    | 1.0000 | 1.0000   |
| 6         | 1.0000    | 1.0000 | 1.0000   | 1.0000    | 1.0000 | 1.0000   |
| 7         | 1.0000    | 1.0000 | 1.0000   | 1.0000    | 1.0000 | 1.0000   |
| 8         | 0.7273    | 1.0000 | 0.8421   | 0.7273    | 1.0000 | 0.8421   |
| 9         | 1.0000    | 1.0000 | 1.0000   | 1.0000    | 1.0000 | 1.0000   |
| Macro Avg | 0.9727    | 0.9625 | 0.9611   | 0.9727    | 0.9625 | 0.9611   |

*GMM-32 values from notebook output. Fill GMM-16 column from your run.*

#### 4.3.4 Overall Comparison: GMM-16 vs GMM-32

Table 5: Overall Performance Comparison

| System             | Accuracy (%) | Macro Precision | Macro Recall  | Macro F1      |
|--------------------|--------------|-----------------|---------------|---------------|
| GMM-16             | <b>96.25</b> | <b>0.9727</b>   | <b>0.9625</b> | <b>0.9611</b> |
| GMM-32             | <b>96.25</b> | <b>0.9727</b>   | <b>0.9625</b> | <b>0.9611</b> |
| $\Delta$ (32 – 16) | 0.00         | 0.00            | 0.00          | 0.00          |

#### 4.4 Discussion

- GMM-16 and GMM-32 achieve identical recognition accuracy, indicating that increasing the number of Gaussian mixtures did not improve classification performance for this dataset.
- The confusion between digit 1 and digit 8 is observed in both models, suggesting acoustic overlap in the MFCC feature space rather than insufficient model complexity.
- Since both systems produce the same confusion matrix, it implies that 16 mixtures are already sufficient to model the acoustic distribution of the spoken digits.
- Increasing the number of mixtures beyond 16 does not provide additional discriminative power and may lead to overfitting given the limited training data (17–18 utterances per digit).
- Further improvement could be achieved using techniques such as CMVN (cepstral mean and variance normalisation) or by increasing the amount of training data.

## 5 Project 3: GMM-HMM and DNN-HMM Based ASR (Kaldi)

### 5.1 Objective

To develop continuous speech recognition (ASR) systems using the Kaldi toolkit, implementing both a context-dependent GMM-HMM baseline and a hybrid DNN-HMM system, and to evaluate their performance using Word Error Rate (WER).

- A context-dependent GMM-HMM system (baseline)
- A hybrid DNN-HMM system

The performance of both systems is evaluated using Word Error Rate (WER).

### 5.2 Dataset and Setup

- Toolkit: Kaldi
- Features: 13-dimensional MFCC with CMVN
- Acoustic Models:
  - GMM-HMM (monophone  $\rightarrow$  triphone)
  - DNN-HMM (feed-forward neural network)
- Language Model: Bigram LM
- Decoding: Viterbi algorithm

### 5.3 Result

Table 6: ASR System Comparison

| System           | WER (%) |
|------------------|---------|
| Monophone HMM    | 68.17   |
| Triphone GMM-HMM | 49.75   |
| DNN-HMM          | 48.52   |

### 5.4 Discussion

- The monophone HMM gives a high WER of 68.17% because it models each phone independently without considering neighboring context.
- The triphone GMM-HMM significantly reduces WER to 49.75% by modelling left and right phonetic context, which better captures coarticulation effects in continuous speech.

- The DNN-HMM achieves the lowest WER of 48.52%, showing a small but consistent improvement over the GMM-HMM baseline. This improvement comes from the DNN’s ability to learn more complex acoustic patterns.
- The improvement from monophone to triphone is large, while the improvement from GMM-HMM to DNN-HMM is smaller, indicating that context modelling contributes more than increasing acoustic model complexity.
- Overall, adding contextual modelling and neural acoustic modelling improves recognition accuracy and reduces word errors.

## 6 Conclusion

- In Project 1, DTW was used for isolated vowel recognition. It worked well for small vocabulary tasks and did not require any training. However, it is computationally expensive and depends heavily on the chosen reference templates.
- In Project 2, GMM was used for isolated digit recognition. Compared to DTW, GMM performed better because it models the statistical distribution of speech features. Increasing the number of mixtures improved performance slightly, but too many mixtures can lead to overfitting.
- In Project 3, GMM-HMM and DNN-HMM were used for continuous speech recognition. The triphone GMM-HMM significantly improved performance compared to monophone models. The DNN-HMM gave the best results among all systems, showing that neural networks can model speech features more effectively.
- Overall, the projects show the progression of speech recognition systems — from simple template matching (DTW), to probabilistic modelling (GMM), and finally to sequence modelling with neural networks (HMM/DNN). Each step improves accuracy and modelling capability.

## 7 Reference

1. L. R. Rabiner, “A Tutorial on Hidden Markov Models and Selected Applications in Speech Recognition,” *Proceedings of the IEEE*, vol. 77, no. 2, pp. 257–286, 1989.
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3. D. A. Reynolds and R. C. Rose, “Robust Text-Independent Speaker Identification Using Gaussian Mixture Speaker Models,” *IEEE Transactions on Speech and Audio Processing*, 1995.